DE–3373 **11**

DISTANCE EDUCATION

P.G.D.C.A. (S) DIPLOMA EXAMINATION, MAY 2012.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) Discuss the various generations of Computers. (10)

   (b) What are the ethics of information technology? Explain. (10)

2. (a) Explain the four types of application software. (10)

   (b) Discuss in detail on Internet web browsers. (10)

3. (a) Explain in detail on Electronic Data Interchange.(10)

   (b) Explain the factors affecting the data transmission. (10)

4. (a) Explain any two types of secondary storage devices. (10)

   (b) What are the features of Data Base Management Systems? Explain. (10)

5. (a) Explain briefly the phases of System Analysis and Design. (10)

(b) Write short notes on Internet Programming. (10)

6. (a) Discuss the various issues of Information Security. (10)

(b) Explain any two types of Database Organisations. (10)

7. (a) Discuss the telephone related communication services. (10)

(b) Write short notes on Intellectual Property Rights. (10)

8. (a) Explain the All Purpose machine of computer communication technology. (10)

(b) Discuss in detail on Multimedia. (10)

DISTANCE EDUCATION

P.G.D.C.A. (S) DEGREE EXAMINATION, MAY 2012.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the hexa decimal number 101.01 in to binary, octal and decimal number systems. (10)

   (b) Perform the following: (10)

   (i) 11011 + 1101 (in binary)

   (ii) 101×110 (in binary)

   (iii) 789 + 292 (in hexa decimal)

2. Simplify the Boolean expression using Karnayh map and implement using NOR gates only.

$$Q(A, B, C, D) = \Sigma m(0, 4, 5, 6, 9, 11, 14, 15)$$

3. (a) Draw the circuit of 4–bit binary parallel adder and explain its working. (12)

   (b) Explain the number representation in the signed 2's complement representation. (8)

4. (a) Draw the circuit of a 4–bit shift register and explain the operations that can be preformed with it. (12)

(b) Explain about RAM and ROM. (8)

5. (a) What is DMA? Explain it with block diagram. Also list its advantages. (12)

(b) Explain the operations in the fetch cycle. (8)

6. (a) Discuss the design of a control unit. (10)

(b) Explain:

(i) Magnetic tape.

(ii) Decade counter. (10)

7. (a) Draw the circuit of a 4×1 multiplexer. (10)

(b) Draw the circuit of 2×4 decoder and explain. (10)

8. Explain:

(a) Instruction cycle. (8)

(b) I_{10} interface. (6)

(c) ASCII code. (6)

DE–3375 **13**

DISTANCE EDUCATION

PGDCA(S) DEGREE EXAMINATION, MAY 2012.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain various data types supported by C.

   (b) Describe the 'gets' and 'puts' functions. (10+10)

2. (a) Explain the 'Branching' statements in C.

   (b) Write a C Program to solve a quadratic equation $ax^2+bx+c=0$. (10+10)

3. (a) Explain the various built-in functions for string manipulations.

   (b) Write C program to calculate a factorial of a integer using function. (10+10)

4. (a) What is 'structure' in C ?

   (b) Write the use of pointers.

   (c) Write a program for file copying. (5+5+10)

5. (a) Write a program to multiply two matrices.

   (b) Write a note on 'Macros and preprocessors' (12+8)

6. (a) Explain the various stack operations.

(b) Write a note on the role of 'Data structures'. (12+8)

7. (a) Explain Binary Tree and its representation.

(b) Explain 'Doubly linked list' with neat diagram. (10+10)

8. (a) Write the applications of Tree.

(b) Write In order, preorder and post order Tree traversals with examples. (5+15)

DE–3376 **14**

DISTANCE EDUCATION

P.G.D.C.A (S) EXAMINATION, MAY 2012.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the working with windows elements.

   (b) How to start and exit office application? Give any FOUR applications of MS-Office.

2. (a) How to creating, opening, saving and save as a files in MS-Word? (8)

   (b) Write short notes on (12)

      (i) Edit and delete the text

      (ii) Moving and copying text

      (iii) Find and Replace.

3. (a) How to create a table in MS-Word? Explain with an example. (8)

   (b) Write short notes on (12)

      (i) Create bullet and numbered list

      (ii) Change margins and change line spacing

      (iii) Page numbering and heading.

4. (a) Explain any five functions in MS-Excel. (10)

(b) Explain any three types of charts in MS-Excel. (10)

5. Write short notes on

(a) Entering data and time (4)

(b) Working with ranges (3)

(c) Editing and deleting entries (4)

(d) Insert and delete rows and columns (4)

(e) Build an excel data base. (5)

6. (a) To create the power point presentation for the college department advertisement with 5 slides.

(b) How to create graph chart and organization chart? Explain.

7. (a) Write down the procedure for the creating and save the table with student mark details.

(b) How to adding, inserting and deleting records with an example.

8. (a) How to creating, saving and editing a query forms? Explain.

(b) How to creating and printing Reports of the employee details? Explain.

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DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What are data types? Mention all predefined data types in C++.

   (b) What do you mean by dynamic initialization of a variable? Give an example.

2. (a) What are the special characteristics of construction function?

   (b) Write a complete object with string constant.

3. (a) List out the rules of overloading operator.

   (b) Write a C++ program to perform 2*2 Matrix addition.

4. (a) What are the unique advantages of an object oriented programming?

   (b) List a few areas of application of OOP technology.

5. (a) Describe with example the uses of enumeration data types.

   (b) What do you mean by default arguments? Discuss with an example.

6. (a) Discuss types of Constructor in detail with an example.

   (b) Explain about file pointer with an example.

7. (a) What is file? Discuss in detail with an example.

   (b) What is command line argument? Discuss with an example.

8. (a) What are the rules for overloading operators in C++.

   (b) Why is it necessary to include the file iostream.h in all our program? Explain with an example.

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Define system. Mention its characteristics.

   (b) List the Inter personal skills needed by a system analyst.

2. (a) Give the difference between a open system and a closed system.

   (b) Discuss on:

      (i) MIS

      (ii) DSS.

3. (a) Mention the need for planning in system analysis.

   (b) Draw the blocks used in Data flow diagram and discuss their purpose.

4. (a) List the steps to be followed in Feasibility Analysis.

   (b) Give the purpose of data dictionary.

5. (a) How logical system design differs from physical system design?

(b) What is the need for Input Validation?

6. (a) How a form is defined?

(b) Mention the requirements of form design.

7. (a) Compare sequential access with direct access in file concepts.

(b) What are the symbols used in system flow chart? Mention their purpose.

8. (a) Mention the procedure for selecting Hardware and Software specification for system development.

(b) Give the importance of personnel training.

DE–3379 **23**

DISTANCE EDUCATION

P.G.D.C.A. (S). EXAMINATION, MAY 2012.

Second Semester

VISUAL PROGRAMMING — I

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain types of forms with its properties and Events.

   (b) Write down the features of Visual Basic.

2. (a) With example explain looping and Control Structure.

   (b) Discuss about Visual Basic data types.

3. (a) Explain any five Visual Basic controls with its properties, methods and events.

   (b) Write briefly about printer object supported by Visual Basic.

4. (a) Explain with example string functions and date functions in VB.

   (b) Give a brief note on List and Combo boxes with its properties, methods and events.

5. (a) List out the uses of functions and procedures.

   (b) Discuss about flex grid controls.

6. (a) Write brief note on object oriented programming.

(b) Explain different steps in creating a Menu Editor.

7. (a) What do you mean by Active X control? How it can be created in VB?

(b) Give a brief note on various data control methods and events?

8. (a) Explain various controls provided by VB for file management.

(b) Write short note on OLE.

DE–3380 **24**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain in detail about DDA Line drawing algorithm. (10)

   (b) Explain the mid point circle generation algorithm. (10)

2. (a) Discuss in detail about viewing transformations. (10)

   (b) Explain the convex polygon clipping with example. (10)

3. (a) Give the matrix representation for three dimensional scaling and rotation transformation. (10)

   (b) Explain in detail about three dimensional translation and mirror reflection. (10)

4. (a) Write in detail about z-buffer algorithm. (10)

   (b) How do you construct a three dimensional view? - Explain. (10)

5. (a) Explain in detail about the importance of information display. (10)

(b) Discuss in detail about user's model. (10)

6. (a) Bringout the significance of point clipping. (10)

(b) What is aspect ratio? (5)

(c) Explain the principles of transformation. (5)

7. (a) Describe in detail about three dimensional viewing process. (10)

(b) Explain in detail about region filling techniques. (10)

8. (a) Write in detail about the importance of command language. (10)

(b) Explain in detail about video display generation. (10)

DE–4338 **15**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

LAB I — DATA STRUCTURES USING C

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the examiner and given to the students by Lot system.

Each subdivision carries 50 marks.

1. (a) Write a menu driven program for matrix manipulations.

   (b) Write a C program to sort a set of elements using bubble sort.

———————————— cut here ————————————

2. (a) Write a C program to implement push and pop operations on stack.

   (b) Write a C program to find an element in an array and its location.

———————————— cut here ————————————

3. (a) Write a C program to implement insert and delete operations on Queue.

   (b) Write a C program to count the number of vowels in a line of text.

4. (a) Write a C program to solve a quadratic equation using function.

(b) Write a C program to sort n numbers using selection sort.

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5. (a) Write a C program to implement all string operations.

(b) Write a menu driven program to find an element using linear and binary search.

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, MAY 2012.

First Semester

COMPUTER LAB — II : MS OFFICE AND INTERNET

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

ONE question should be selected by the Examiner and given to the candidates by lot system.

Each sub-division carries 50 marks.

1. (a) Create a new document and type the following equation:

$$f(x) = \alpha_0 + \sum_{n=1}^{\infty} (\alpha_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}$$

(b) Create HTML page to display the pictures of Elephant and monkey. To link the page when user clicks over the picture. The linked page should display features of the animal clicked.

2. (a) Create a new document and type the following equation:

$$f(x) = \alpha_0 + \sum_{n=1}^{\infty} (\alpha_n \cos n\pi x + b_n \sin n\pi x)$$

(b) Create the following inventory Worksheet in MS-Excel.

ITEM No	101
Name	BOLTS
Quality	2000
Stock	500
Reorder	300
Purchased	560
Issued on hand	200

Quantity Level

(i) Enter all the data item except QUANTITY ON HAND for 10 items

(ii) Find QUANTITY ON HAND using the formula.

STOCK QUANTITY = QUANTITY ON HAND + QUANTITY PURCHASED – QUANTITY ISSUED

Find total stock value in inventory as a product of total quantity hand and total price ($\sum$ quantity on hand * $\sum$ price). Display all the item in red color whose quantity on hand is below reorder level.

3. (a) Create a form letter that informs the customer about the data of maturity of a deposit amount in a commercial bank and request the customer for renewal. Mail merge it with an access database containing all the customer data. Prepare letters for customer whose due data falls in a specific range of dates.

(b) Create HTML page to display 10 course and their highlighted features, conducted by an educational institution in a colorful format.

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4. (a) Create a table of student data that contains REGNO, NAME, TAMIL, ENGLISH and MATHS marks. Add a new column named TOTAL and find the row total for each student.

Add two rows named TOTAL, AVERAGE. Find the total and average values for each subject mark. Convert the table to text.

(b) Create HTML page that divides the screen space into three frames (one vertical and two horizontal of equal size). Display three different files in them with scrolling titles relevant to their contents.

5. (a) Create the following worksheet in EXCEL for Electricity Bill

CONSUMER NO	TYPE	NAME	PMR	CMR
101	D	RAVI	580	720

(i) Add data for 10 customers with type 'D' for domestic user and type 'I' for industrial user.

(ii) Find UNITS column for each customer and calculate bill using the slab given below.

(iii) All the customers should be charges a minimum bill of Rs.20% = even of their bill amount is below Rs.20/=

TYPE D CONSUMER

UNITS CONSUMED	RATE/UNIT Rs.
FIRST 100	0.75
NEXT 50	1.25
REMAINING	2.00

TYPE I CONSUMER

UNITS CONSUMED	RATE/UNIT Rs.
FIRST 100	1.10
NEXT 900	2.40
REMAINING	3.50

(b) Create HTML page that plays video and audio files.

DE–4340 **25**

DISTANCE EDUCATION

P.G.D.C.A. (Semester) EXAMINATION, MAY 2012.

Second Semester

COMPUTER LAB III — GRAPHICS USING C++

(2005 Upto 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

ONE question should be selected by the examiner and given to the candidate by lot system.

Each question carries 100 marks.

1. Write a C++ program to draw an ellipse using midpoint ellipse drawing algorithm.

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2. Write a C++ program to draw a triangle ABC, and rotate it both in clockwise and anticlockwise direction.

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3. Write a C++ program to implement the Sutherland-Hodgeman Polygon clipping algorithm.

4. Write a C++ program to draw a circle using Bresenham's circle algorithm.

---------- Cut here ----------

5. Write a C++ program to manipulate an object using shading technique.

---------- Cut here ----------

6. Write a C++ program to draw any 3-D object and perform animation on it.

---------- Cut here ----------

7. Write a C++ program to draw a cube and rotate it along x-axis, y-axis and z-axis.

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8. Write a C++ program to draw different kinds of thick lines using Bresenham's line algorithm.

DE–4341 **26**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

VISUAL BASIC (COMPUTER LAB IV)

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Break-up of mark

Record Note Book :	5
Algorithm/Flow chart :	15
Program :	35
Debugging and Execution :	35
Result :	10
Total :	100

One question should be selected by the Examiner and given to the candidate by lot system.

1. Write a VB Program to calculate simple interest or compound interest using three command Buttons namely Input, Calculate, Display.

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2. Write a VB project that stores 10 students records with fields SREGNO, NAME, MARK in an array. Display data fields in test boxes and provide command buttons to move to desired record.

3. Write a program using control array that creates a scientific calculator with appropriate command buttons.

---------- Cut here ----------

4. Write a VB application to create a list box and do the following operations to the list box:

   (a) Add entries to the list box

   (b) Delete entries from the list box

   (c) Count the number of entries in the list box

   (d) Clear the list box

---------- Cut here ----------

5. Design a VB application to generate a pay bill for an employee in an organization.

---------- Cut here ----------

6. Write a VB program using Active X control to create a text box that accepts only numeric value. Provide the following properties for the text box: Backcolor, Forecolor and Text.

---------- Cut here ----------

7. Develop a procedure to create a text editor which includes the options such as cut and paste operations of selected text and searching a required word in a file.

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8. Write a VB project using frame, check box, text box, option button, command button to enter employee data. Add the data into an array.

9. Write a VB project for commercial bank operations using SB account database, with the following features.

   (a) ADD NEW ACCOUNT

   (b) DEPOSIT AMOUNT

   (c) WITHDRAW AMOUNT

   (d) CLOSE ACCOUNT

---------- Cut here ----------

10. Write a VB application that would receive the name of a shape and supporting parameters from the user and tender the shape on the form.